

S-SPARC: Retrieval-first AI for responsible use in programming education

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Abstract

Educational artificial intelligence (AI) research has paid limited attention to how routine large language model (LLM) use shapes students' awareness of computational resource consumption. S-SPARC addresses this educational concern through a retrieval-first programming assistant that reuses validated responses before invoking external LLM inference and provides resource feedback with gamified regulation. The study comprised retrieval calibration with 200 human-labeled prompts, a within-participant comparison involving 20 students using GPT-5.2, and a seven-session deployment involving 55 students. In the within-participant comparison, average token use decreased from 8,525.15 to 1,980.20 tokens per student, a significant reduction of approximately 77% ($t(19) = 2.36$, $p = 0.029$, Cohen's $d = 0.53$). Mean code-quality scores were 87.72% and 88.13%, respectively, and are interpreted descriptively. Across seven sessions, retrieval handled 1,781,845 of 2,122,873 potential tokens, reducing external LLM inference demand by 83.94%. Survey findings provide descriptive evidence about student perceptions, while the environmental indicators represent proxies rather than direct measurements. These findings demonstrate the potential of validated retrieval to reduce external LLM inference demand and support responsible AI use in programming education.

Keywords: sustainable AI, programming education, responsible AI use, semantic caching, LLM inference reduction, gamification, environmental footprint

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1. Introduction

A programming assistant based on a large language model (LLM) that always generates a new answer teaches a quiet lesson: computational resources can be ignored. This lesson is increasingly difficult to justify. Students often ask variations of the same debugging, syntax, or algorithmic question. Conventional LLM workflows invoke new external LLM inference even when a validated answer to a similar problem may already exist. The educational concern is therefore not only whether artificial intelligence (AI) produces useful answers, but also which habits of dependence, selectivity, and resource use the learning environment encourages.

Educational chatbot research reports benefits for support, usability, and engagement [1]. Broader reviews are also available [2]. Retrieval-augmented generation (RAG) has extended this work by grounding educational responses in external knowledge and examining retrieval quality in educational chatbots [3]. Studies in programming and software engineering education further show how retrieval can provide domain-specific support for coding tasks [4, 5]. Broader surveys confirm that educational RAG is an active and diverse area rather than a single system pattern [6, 7]. S-SPARC builds on this established educational use of retrieval; it does not claim retrieval-supported programming assistance as a new idea.

Sustainable AI research provides the resource-accounting motivation for the study. Green AI argues that computational demand should be considered alongside task performance [8]. Green Algorithms provides a general approach to carbon-footprint estimation [9], while inference benchmarking examines energy, carbon, and water demand for LLM use [10]. S-SPARC does not introduce a new environmental-accounting method. It uses indicators based on token counts and application programming interface (API) outputs to make resource demand visible within an educational interface, subject to the measurement limitations reported later in the manuscript.

Semantic caching provides the closest technical foundation. GPTCache demonstrates reuse for cost and latency reduction [11]; SCALM studies semantic caching for automated chat services [12]; ContextCache addresses multi-turn context [13]; and MeanCache adopts a user-centered perspective [14]. Other work examines domain-specific embeddings and synthetic data for semantic caching [15], while verified semantic prompt caching shows that

validation is also an emerging concern in this area [16]. Taken together, these studies show that semantic reuse, user-centered caching, embedding adaptation, and verification are not unique to S-SPARC. The present study instead applies these ideas in a programming education workflow that attempts retrieval before external LLM inference and exposes resource-use indicators to learners.

Gamification is likewise established in education. Meta-analytic evidence indicates that its effects depend on the instructional context and design choices [17]. Leaderboards have been studied in engineering education [18], and gamified designs have been evaluated in introductory programming and mastery-learning settings [19, 20]. S-SPARC therefore does not claim gamification as a novel educational mechanism. Its design choice is to align leaderboard scoring with relative token efficiency rather than participation volume, although the present study does not isolate the effect of that choice.

Taken together, the cited literature supplies all of the main components used by S-SPARC: educational retrieval, semantic reuse, verification, footprint estimation, and gamification. The contribution of this study is therefore not the invention of any individual component. Its more limited position is the integration and classroom evaluation of these components around a specific design objective: reducing external LLM demand while making resource use visible in programming education. Because the evaluation does not include component-level ablations or a separate behavioral control condition, the study cannot establish that this particular combination is superior to other retrieval, caching, or gamified educational designs.

This study addresses three research questions. Research question 1 (RQ1) asks how external LLM inference demand differs between the direct ChatGPT and S-SPARC workflows and what cumulative reduction is recorded during course deployment. Research question 2 (RQ2) asks how code-quality scores compare across the two workflows. Research question 3 (RQ3) asks what pre-use AI-use patterns, pre- and post-use sustainability response distributions, and post-use perceptions are reported in the available survey data. These questions are examined through a working programming-assistance system, a within-participant comparison with a baseline ChatGPT workflow, and a semester-long quasi-experiment.

The paper makes four contributions. First, it applies responsible AI use as an educational design objective concerned with selective use of generative assistance. Second, it documents a context-specific architecture that combines established retrieval, validation, footprint feedback, and gamification

mechanisms. Third, it reports token reduction and descriptive code-quality results from two programming education settings. Fourth, it reports survey distributions and post-use perceptions while explicitly avoiding attribution of those observations to individual intervention components.

2. Conceptual background and research gap

2.1. Educational AI and retrieval-supported learning

Research on educational chatbots has developed from general conversational support toward retrieval-grounded assistance [2, 3]. Educational RAG studies and surveys address grounding, relevance, reliability, implementation patterns, and domain adaptation [6, 7]. In programming and software engineering education, retrieval has been used to support coding and technical tasks [4, 5]. S-SPARC belongs within this educational RAG tradition. Its evaluation adds a resource-demand outcome, but the available evidence does not show that this outcome has been absent from every prior educational retrieval system.

2.2. Semantic caching and validated retrieval

Semantic caching directly targets repeated LLM calls. GPTRCache emphasizes faster responses and cost savings [11]; SCALM applies semantic caching to automated chat services [12]; ContextCache considers multi-turn queries [13]; and MeanCache emphasizes user-centered web services [14]. Research also examines domain-specific representations for caching [15] and verified semantic prompt caching [16]. This body of work substantially overlaps with S-SPARC’s reuse and validation mechanisms. The distinguishing context of S-SPARC is educational deployment and learner-facing feedback, not a new caching or verification algorithm.

Retrieval quality depends on representation and indexing. Sentence-BERT supports sentence-level semantic comparison [21]. Language-agnostic embeddings support cross-language representation [22]. Facebook AI Similarity Search (FAISS) supports scalable similarity search [23]. S-SPARC uses these established tools as implementation components. No comparative claim is made for concatenating the selected embeddings because alternative representations were not evaluated.

2.3. Sustainable AI and behavioral regulation

The environmental motivation comes from Green AI and studies of computational footprints. Verdecchia et al. describe the broader Green AI agenda [8]; Green Algorithms provides a method for carbon-footprint estimation [9]; and LLM inference studies report energy, carbon, and water implications [10]. S-SPARC draws on this literature but uses proxy indicators rather than direct hardware measurement. Its contribution is limited to placing those indicators alongside a retrieval-first educational workflow.

Gamification research provides the basis for the leaderboard and scoring design. A meta-analysis emphasizes the importance of instructional context [17], while studies in engineering and programming education demonstrate established uses of leaderboards and other game elements [18, 19, 20]. S-SPARC adapts these ideas to token-efficiency feedback. Since no gamification-only condition was evaluated, the study reports this as a design adaptation rather than evidence of a distinct behavioral effect.

2.4. Research gap and study positioning

The cited literature shows that retrieval, semantic caching, verification, footprint estimation, and gamification are all established research directions. S-SPARC is positioned as a context-specific integration of these directions rather than as the first system to combine efficiency and educational support. The study asks whether this integrated workflow can reduce recorded external LLM demand in programming courses while providing descriptive evidence about output quality and learner perceptions. Claims about behavioral change, component-level effectiveness, and superiority over other caching or educational RAG systems remain outside the available evidence.

Table 1: Relationship between the cited research streams and S-SPARC

Research stream	Contribution established in cited work	How S-SPARC uses the stream
Educational chat-bots and RAG	Conversational support, grounding, relevance, and domain-specific assistance [2, 3, 4]	Applies retrieval-supported assistance in programming education
Semantic caching	Reuse for latency, cost, context, and user-centered service design [11, 12, 13, 14]	Uses retrieval before external generation to reduce repeated calls
Verified and adapted caching	Domain-specific representations and verification of reusable prompts [15, 16]	Applies multi-embedding retrieval and validation to reusable programming solutions
Green AI and footprint accounting	Resource-awareness and environmental estimation [8, 9, 10]	Displays proxy footprint indicators and reports token demand
Gamified learning	Context-dependent use of leaderboards and game elements [17, 18, 19]	Adapts scoring toward relative token efficiency

3. System design and methodology

3.1. Design principles and rationale

The system design was guided by four principles. **Reuse before generation** means that validated local knowledge is checked before external LLM inference is used. **Validate before reuse** means that no generated answer becomes reusable until it passes multi-metric validation. **Awareness before optimization** requires the environmental footprint to be visible to students before behavior can be expected to change. **Behavior before efficiency** means that the system optimizes how students choose to prompt, not only raw token savings.

These principles guide the architecture described in the following subsections.

S-SPARC follows a conservative rule: external LLM inference is used only when validated local knowledge cannot answer the query. This rule is stricter than ordinary semantic caching because careless reuse is particularly risky in programming education. A cached answer may be semantically similar yet still be incorrect for a student’s problem. Retrieval, validation, footprint feedback, and gamification were therefore designed as connected components rather than optional features.

The system has four layers. The retrieval layer searches for reusable programming knowledge. The validation layer controls which generated prompt–solution pairs may enter the validated knowledge base. The footprint feedback layer converts token and application programming interface (API) usage into approximate environmental-footprint estimates. The gamification layer makes efficient use a visible behavioral target. Together, these layers make responsible consumption a system behavior that students can observe and act upon.

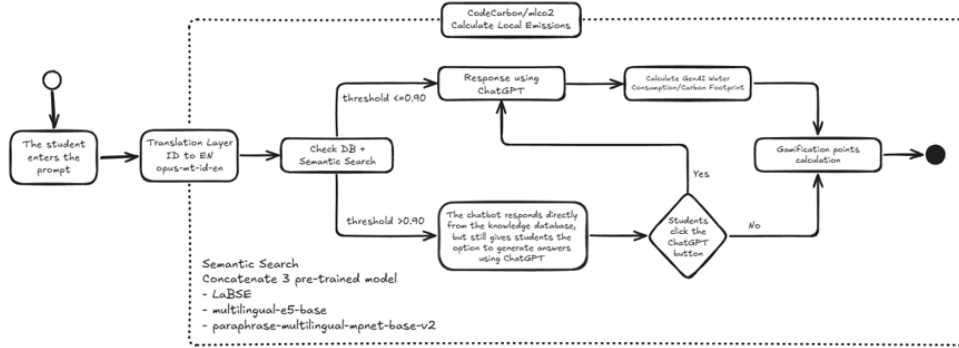


Figure 1: S-SPARC operational workflow. The system logs each query, prioritizes validated retrieval, and invokes external LLM inference only when reuse is not justified.

3.2. Prototype implementation and logging

The thesis describes a Python prototype with a Flask backend, a BotMan interface, and a web interface. The system used the OpenAI API for external generation and logged prompts, chat history, inference-job metadata, response latency, token use, cache hits and misses, queries, and responses. The controlled comparison used GPT-5.2. The available project files do not preserve the software-library versions, OpenAI API configuration, generation

Algorithm 1 Request processing pipeline

Require: Student query q , validated knowledge base K , threshold $\tau = 0.90$ calibrated from available MBPP and HumanEval data

- 1: Log q , chat history, and inference metadata
 - 2: Detect language and translate q when required
 - 3: Compute the concatenated embedding e_q
 - 4: Retrieve nearest candidates from K using FAISS
 - 5: **if** top candidate similarity $\geq \tau$ and candidate is validated **then**
 - 6: Return the reusable solution and record retrieval-based reduction
 - 7: **else**
 - 8: Invoke the external LLM and record token usage
 - 9: Send the generated prompt–solution pair to validation
 - 10: **end if**
 - 11: Update footprint and gamification feedback
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parameters, or the model version used for every interaction in the semester-long deployment. These details are therefore not specified beyond what is recorded in the thesis.

3.3. Knowledge sources and query representation

The initial knowledge base was derived from Mostly Basic Python Problems (MBPP) and HumanEval. MBPP contains 974 elementary programming problems with unit tests, while HumanEval contains 164 programming problems with function signatures and test cases. The benchmark items remained in English to preserve their original structure. Students could submit queries in Indonesian; the system automatically translated these queries into English before retrieval. The translation service and model version remain unspecified, so translation quality could not be reproduced or evaluated independently.

Each interaction is logged with its prompt, chat history, and inference-job metadata. This record supports later analysis of token use and allows newly generated answers to be scored before they become candidates for reusable knowledge. The thesis specifies the scoring procedure but does not preserve the final numeric admission cutoff. The manuscript therefore reports the validation score without claiming that the final admission decision can be independently reproduced from the available files.

3.4. Retrieval, generation, and knowledge validation

Prompt representations were built using three pretrained multilingual embedding models. These were Language-agnostic BERT Sentence Embedding (LaBSE), multilingual-e5-large, and paraphrase-multilingual-mpnet-base-v2. Their vectors were concatenated before indexing with FAISS. The thesis describes concatenation rather than learned or manually assigned fusion weights. Queries and stored prompts were compared using cosine similarity. The operational retrieval threshold was set at 0.90 to prioritize precision over aggressive reuse, based on calibration using the available MBPP (974 problems) and HumanEval (164 problems) benchmark data. Because no independent held-out threshold-selection set was used, generalization beyond the scale and coverage of these benchmarks remains uncertain. Exact model revisions and library versions were not preserved in the available files.

Candidate prompt–solution pairs were scored using five metrics: prompt–code alignment (A), logical quality (L), semantic similarity to a reference solution (S), code quality (Q), and static-analysis score (S_s). For a prompt P , generated code C , and reference code C_{ref} , the thesis defines

$$A = 10 \text{ sim}(P, C), \quad (1)$$

$$L = 3V + 0.55S_s + B, \quad (2)$$

$$S = 10 \text{ sim}(C, C_{\text{ref}}), \quad (3)$$

$$Q = 0.35A + 0.55S_s + I, \quad (4)$$

$$F = 0.40A + 0.25L + 0.20S + 0.10Q + 0.05S_s, \quad (5)$$

where cosine similarity is scaled to $[0, 10]$, S_s is a static-analysis score scaled to $[0, 10]$, $V = 1$ for valid syntax and 0.5 otherwise, and $I = 1$ for valid syntax and 0 otherwise. The length bonus is $B = 1$ for 1–12 lines of code, $B = 0.4$ for more than 12 lines, and $B = 0$ when no code is present. The final score F is bounded to $[0, 10]$. The thesis states that the weights were empirically calibrated through grid search to align the system score with human judgment, with 65% of the composite weight assigned to alignment and logic. However, the calibration data, human-rating protocol, static-analysis implementation, and final admission cutoff are not available in the project files. These components are therefore reported as prototype procedures rather than independently reproducible validation guarantees.

Table 2: Retrieval and validation configuration for S-SPARC

Component	Configuration or role
Input normalization	Indonesian queries are detected and translated before retrieval
Representation	LaBSE, multilingual-e5-large, and paraphrase-multilingual-mpnet-base-v2
Fusion	Vector concatenation before indexing
Indexing	FAISS nearest-neighbor search with cosine-similarity interpretation
Operational threshold	0.90, selected from MBPP and HumanEval calibration data to prioritize precision; not independently validated on a held-out set
Knowledge scoring	Composite score F based on alignment, logic, semantic similarity, code quality, and static analysis
Unavailable configuration	Model revisions, library versions, translation model, static-analysis tool, and final admission cutoff

Algorithm 2 Knowledge validation workflow

Require: Candidate prompt–solution pair (q, a)

- 1: Estimate prompt–solution alignment
 - 2: Evaluate logical correctness of the proposed solution
 - 3: Compute semantic-similarity and code-quality indicators
 - 4: Apply static-analysis signals where applicable
 - 5: Combine validation metrics, emphasizing alignment and logic
 - 6: **if** combined validation score satisfies the admission policy **then**
 - 7: Add (q, a) and metadata to the reusable knowledge base
 - 8: **else**
 - 9: Exclude (q, a) from future retrieval reuse
 - 10: **end if**
-

3.5. Environmental-footprint accounting

The thesis adopts an inference-energy framework from Jegham et al. in which per-query energy depends on output length, token-generation rate, latency, GPU and non-GPU power, utilization, and power usage effectiveness (PUE) [10]. This framework describes the physical quantities that would be required for hardware-aware accounting. However, S-SPARC accessed the LLM through a managed cloud service. GPU power, node-level utilization, internal token-generation rate, and data-center overhead were therefore not directly observable in this study.

The implemented accounting consequently used token counts as a proxy for inference activity. As documented in the thesis, the number of processed tokens was combined with an average energy-per-token estimate from the literature. The resulting energy estimate was then used to derive carbon and water indicators through carbon-intensity, PUE, and water-usage factors. The available thesis and project files do not preserve the energy-per-token value, carbon-intensity value used by the environmental API, PUE value, water-usage factor, API name, API version, or configuration. The physical conversion cannot therefore be independently reproduced from the available materials.

The system logged token use and displayed the returned footprint indicators at interaction and session levels. The thesis reports average outputs from the environmental API of approximately 0.000503 kWh, 0.0152 kg CO₂, and 0.19 L of water per execution. It separately reports approximately 26.3 kg CO₂ for one controlled laboratory session and expresses this value as ap-

proximately 34 kWh using an Indonesian grid-emission factor of 0.8 kg CO₂ per kWh. The 34 kWh value is a carbon-to-electricity equivalence derived from the reported carbon value, not a direct electricity measurement.

The available files do not document the number of executions included in the session aggregate or the complete aggregation procedure linking the average per-execution outputs to the session total. The per-execution indicators and session-level value therefore cannot be reconciled independently. They are retained as values reported by the thesis, but they are not interpreted as measured hardware consumption, verified emissions, or environmental savings caused by S-SPARC. The directly observed sustainability-related quantity is token demand, including the number of tokens handled through retrieval rather than external LLM inference.

3.6. Gamified regulation

Gamification used a dynamic threshold based on group token use. During each evaluation period, the threshold was set to 110% of the participants' mean token use. Participants at or below the threshold received the maximum efficiency score, while scores above the threshold were reduced proportionally. The detailed formula was not disclosed to participants during the experiment to reduce strategic manipulation. Scores were displayed through a leaderboard, and the highest-scoring participants received recognition. The thesis does not provide the exact proportional penalty function or a separate experimental condition for gamification. Survey responses concerning gamification are therefore treated as post-use perceptions of the combined system and cannot establish an independent gamification effect.

4. Evaluation design and measures

4.1. Evaluation questions and retrieval assessment

The evaluation addressed four methodological questions. First, what retrieval performance is observed during prototype calibration? Second, how do token use and code-quality scores compare across the two implemented workflows? Third, how does the reduction rate vary during semester-long use? Fourth, what pre-use AI-use patterns, pre- and post-use response distributions, and post-use perceptions are documented in the survey data?

Retrieval quality was evaluated using human labels and weak labels. The human-labeled set contained 200 prompts and 20 candidate documents per prompt, producing 4,000 pairwise relevance judgments. The threshold results

Table 3: Scope of the environmental-footprint accounting. These values are proxy indicators derived from token counts and literature-based energy estimates; they are not direct hardware measurements and cannot be independently reconciled with session-level totals.

Accounting element	Documented use	Interpretation limit
Token logs	Observed inference activity for each workflow	Proxy for external inference demand, not energy measurement
Hardware-aware parameters	Methodological framework includes latency, generation rate, power, utilization, and PUE	Not observable through the managed cloud service
Environmental conversion	Token-based energy estimate converted to carbon and water indicators	Conversion values and API configuration are unavailable
Per-execution outputs	Approximately 0.000503 kWh, 0.0152 kg CO ₂ , and 0.19 L on average	Proxy indicators, not independently verified measurements
Session output	Approximately 26.3 kg CO ₂ ; reported as approximately 34 kWh using 0.8 kg CO ₂ /kWh	Carbon-derived equivalence; aggregation procedure is unavailable

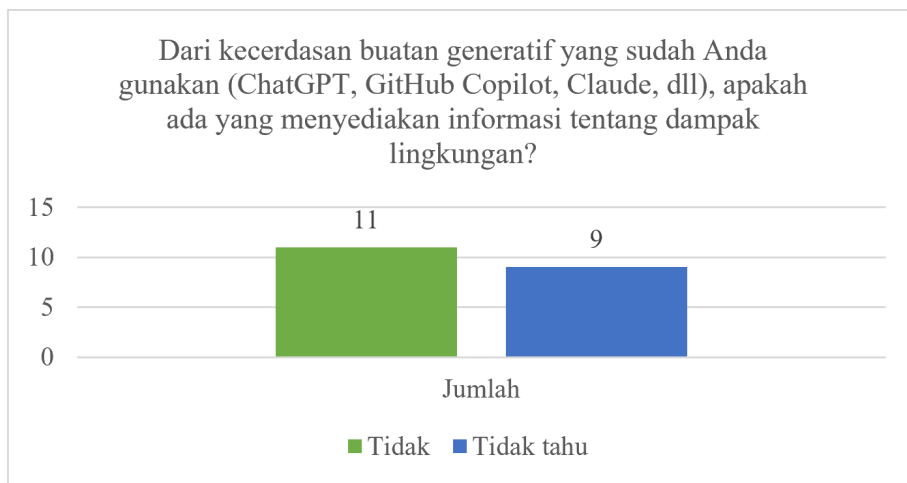


Figure 2: Pre-intervention views on environmental-impact transparency. The figure summarizes student views on displaying footprint information in educational AI systems. The footprint values referenced by the interface are proxy indicators derived from token counts and literature-based energy estimates; they are not direct hardware measurements and cannot be independently reconciled with session-level totals.

in the thesis are summarized at the prompt level, but the available files do not describe how the pairwise judgments were reduced to one outcome per prompt. They also do not identify the number of annotators, the annotation rubric, or inter-rater agreement. The 632-prompt weak-label set was generated automatically through system heuristics and embedding-based criteria. Because the weak labels share signals with the retrieval method, they are treated as exploratory rather than independent ground truth.

Thresholds 0.80 and 0.90 were inspected on these same labeled sets derived from the available MBPP (974 problems) and HumanEval (164 problems) benchmark sources; no separate held-out threshold-selection set is documented. The retrieval analysis should therefore be interpreted as calibration evidence for the prototype rather than an independent estimate of production generalization. The operational threshold of 0.90 was selected conservatively because false-positive reuse was considered more harmful than a false negative that triggered external generation, while its generalizability remains bounded by the size and coverage of the two benchmark datasets.

4.2. Controlled comparison

4.2.1. Design and participants

The controlled comparison was conducted on February 4, 2026, from 09:00 to 11:00 Western Indonesian Time in the Advanced Programming 1 laboratory. The two-hour session involved 20 students from the 2023–2024 cohorts: 13 from Informatics and 7 from Information Systems. The thesis does not document the recruitment procedure, eligibility criteria, prior AI-use requirements, demographic characteristics beyond program and cohort, or a sample-size calculation. The sample is therefore described as the group that participated in the documented laboratory session rather than as a representative sample of programming students.

The detailed protocol and appendix provide participant-level data for a paired, within-participant design. Every participant completed the same tasks with both workflows. An earlier planning overview in the thesis describes random class-level allocation involving first-year Basic Programming students, but no separate randomized groups appear in the executed protocol or appendix. The present manuscript therefore follows the detailed implementation record and does not describe the comparison as randomized, between-group, or causal.

4.2.2. Baseline and intervention workflows

The baseline was direct ChatGPT generation using GPT-5.2. It did not use the S-SPARC retrieval, reusable knowledge-base, footprint-feedback, or gamification workflow. The intervention condition used S-SPARC with the same documented external model, GPT-5.2, when retrieval did not provide an answer. Thus, the comparison held the named external model constant but compared two different user workflows rather than two interfaces that differed only in retrieval.

The thesis notes an additional interface difference. Direct ChatGPT accepted multimodal inputs, and some participants submitted complete task descriptions or text derived from images. S-SPARC processed relevant text only. This difference may have affected prompt length and token use. The thesis does not document a common prompt template, standardized input length, or restrictions that made the submitted prompts equivalent across workflows.

4.2.3. Tasks and procedure

Each participant completed three programming tasks: Trapping Rain Water, Word Frequency, and Point of Sales. The thesis describes the tasks as representing basic, intermediate, and advanced difficulty, but it does not document an independent difficulty-validation procedure. All participants completed the same three tasks in both workflows during the laboratory session. The appendix contains a score for each task, an average correctness percentage, and total token use for each participant under each workflow.

The thesis does not record the order in which the two workflows were used, whether the order was counterbalanced, how the two-hour session was divided between conditions, whether participants could revise earlier answers, or whether a washout period was used. Repeated exposure to the same tasks may therefore have introduced order, practice, or carryover effects.

4.2.4. Outcomes and analysis

The documented comparison used token consumption as the measure of resource demand and programming-solution quality as the measure of educational output. Token totals were recorded for each participant and workflow. The participant-level appendix supports a paired analysis because the same respondent identifiers appear under both workflows. The paired token-use test therefore used 20 matched observations. The thesis does not report a normality diagnostic for the paired differences or another assumption check. This omission is relevant because the baseline token values include several extreme observations.

Programming quality was represented by task scores and an average correctness percentage. The thesis describes the scoring dimensions as algorithmic correctness, suitability of programming logic, and correctness of the final result. However, it does not identify the raters, scoring rubric anchors, score-combination rule, blinding procedure, or inter-rater reliability. It also does not report a direct paired quality test, a prespecified equivalence margin, or a non-inferiority analysis. Quality findings are therefore reported descriptively rather than as proof that the workflows were statistically or educationally equivalent.

The thesis planning overview also lists pre-test and post-test learning gains, motivation, self-regulation, academic integrity, analysis of covariance, and mixed-effects analysis. The available results and appendices do not provide the group structure, variables, or model outputs required to verify that these planned analyses were completed. They are therefore not presented as

outcomes of the controlled comparison in this manuscript.

4.3. Semester-long deployment and survey

4.3.1. Design, participants, and setting

The semester-long study used a quasi-experimental observational design with an existing class rather than random assignment. It involved 55 fourth-semester Informatics students enrolled in a Machine Intelligence course. The thesis states that this class was selected because the course was relevant to the use of AI technologies for programming problems. It does not describe additional recruitment criteria, exclusions, attrition, or a sample-size calculation.

The deployment ran from February to April 2026 across seven laboratory sessions. The course materials increased from basic concepts to more complex implementations. All recorded participants used S-SPARC in the authentic course setting. No concurrent control group or alternative-system condition is documented for this phase. Consequently, the semester-long study describes system use and changes over time but cannot estimate how outcomes would have developed without S-SPARC.

4.3.2. Recorded measures and analysis

The documented variables were token-use efficiency, prompt-quality score, response quality, retrieval-token total, external-inference-token total, reduction rate, and indicators related to gamified use. The thesis does not provide the scoring rubric for prompt quality or response quality in this phase. The manuscript therefore reproduces the reported regression coefficients without extending the interpretation beyond the documented associations.

A paired-sample test compared the early and late reduction-rate phases, while repeated-measures analysis of variance (ANOVA) examined variation across the seven sessions. The thesis reports 55 observations in its regression summaries. However, the raw interaction-level data, the definition of the early and late phase boundaries, the nesting structure, the ANOVA degrees of freedom, and the Greenhouse–Geisser correction coefficient are not available in the project files. The reported analyses are therefore presented as thesis results without independent reconstruction or multilevel reanalysis. Non-significant time comparisons are not interpreted as equivalence or proof of stability.

4.3.3. *Survey procedure and scope*

The thesis includes pre-intervention and post-intervention survey appendices with 20 respondent columns and five-point responses. The items covered environmental concern, knowledge of AI energy use and carbon footprint, prompt practices, semantic reuse, power usage effectiveness, perceptions of gamification, and intentions to use AI more selectively. The available files do not document the survey administration dates, response rate, matching procedure, anonymity procedure, or whether all survey respondents were drawn from the controlled comparison, the semester-long deployment, or both.

The pre- and post-surveys were not identical. The post-survey added items specific to the S-SPARC experience, including retrieval, leaderboard use, transparency, and perceived behavioral change. No validated scale structure, reliability coefficient, matched composite score, or inferential pre-post analysis is documented. Survey findings are therefore reported as item-level descriptive values. They do not provide a controlled estimate of behavioral change or isolate the effects of gamification and footprint feedback.

5. Results

5.1. *Retrieval performance and reliability*

This subsection reports the retrieval results for RQ1.

On the human-labeled set, a threshold of 0.80 produced 200 true positives and no false positives or false negatives. At the operational threshold of 0.90, calibrated from the available MBPP and HumanEval benchmark data rather than an independent held-out set, the system produced 192 true positives and 8 false negatives, with no false positives. Precision therefore remained at 100.00%, while recall and retrieval rate were 96.00%. On the weak-label set, threshold 0.80 produced 611 true positives, 20 false positives, no true negatives, and 1 false negative. At threshold 0.90, it produced 580 true positives, 19 false positives, 1 true negative, and 32 false negatives. These values describe performance within benchmark-grounded calibration data whose generalization is limited by the size and coverage of MBPP and HumanEval.

The thesis also reports ranking metrics for the human and weak labels. Both sets achieved Hit@10 of 100%. Precision@10 was 97.60% for the human labels and 29.68% for the weak labels; nDCG@10 was 1.00 and 0.982,

Table 4: Evaluation settings

Evaluation component	Scale	Purpose
Human retrieval labels	200 queries; 20 candidates each; 4,000 judgments	Prompt-level retrieval calibration using human judgments
Weak retrieval labels	632 automatically labeled prompts	Broader stress test under noisier relevance labels
Controlled comparison	20 students; three tasks; GPT-5.2; two-hour session	Paired comparison of direct-generation and S-SPARC workflows; order not documented
Semester-long quasi-experiment	55 students; seven laboratory sessions; February–April 2026	Longitudinal description of cumulative reduction and variation over time
Survey component	20 respondent columns; five-point pre- and post-use items	Descriptive AI-use patterns, sustainability responses, and post-use perceptions

Table 5: Semantic retrieval performance on the benchmark-grounded calibration sets

Label set	Threshold	TP	FP	TN	FN	Precision	Recall	F1 score	Accuracy	Retrieval rate
Human	0.80	200	0	0	0	100.00%	100.00%	100.00%	100.00%	100.00%
Human	0.90	192	0	0	8	100.00%	96.00%	97.96%	96.00%	96.00%
Weak	0.80	611	20	0	1	96.83%	99.84%	98.31%	96.68%	99.84%
Weak	0.90	580	19	1	32	96.83%	94.77%	95.79%	91.93%	94.78%

respectively, and mean reciprocal rank was 1.00 and 0.979. Because label construction, threshold calibration, and evaluation were grounded in the available MBPP and HumanEval data rather than an independent held-out set, these results describe prototype calibration rather than held-out generalization; their external applicability is limited by the scale and coverage of those benchmarks.

Table 6: Ranking metrics reported for the retrieval evaluation

Metric	Weak labels	Human labels
Hit@10	100.00%	100.00%
Precision@10	29.68%	97.60%
nDCG@10	0.982	1.000
Mean reciprocal rank	0.979	1.000

5.2. Token use and code quality in the controlled comparison

This subsection reports the within-participant comparison results for RQ1 and RQ2.

Token use was the outcome subjected to the documented paired significance test. Average token use was 8,525.15 tokens per student in the baseline workflow and 1,980.20 tokens with S-SPARC. The corresponding sample standard deviations reported in the thesis were 12,467.56 and 648.80 tokens. The mean paired difference was 6,544.95 tokens, corresponding to a reduction of 76.77%, reported in the thesis as approximately 77%. For this token-use comparison, the thesis reports $t(19) = 2.36$, $p = 0.029$, and Cohen’s $d = 0.53$. No confidence interval for the paired token difference or assumption diagnostic is reported.

Code quality is presented separately as a descriptive summary. The thesis reports an average score of 87.72% for the baseline and 88.13% for S-SPARC, corresponding descriptively to a difference of 0.41 percentage points. It also reports an S-SPARC standard deviation of 13.38 and a 95% interval of 88.13 ± 5.86 . This interval describes uncertainty around the S-SPARC mean only; it is not a paired statistical test of the difference between workflows. A paired quality-test result, if subsequently calculated from the appendix data, may be inserted here: `{CODE_QUALITY_TEST_RESULT}`. Until such a result is available, the quality comparison should be read as descriptive and not as evidence of a statistically significant difference, equivalence, or non-inferiority.

Table 7: Within-participant token-use results and descriptive code-quality summary for the baseline ChatGPT workflow and S-SPARC. The 95% interval reported here describes uncertainty around the S-SPARC mean only; it is not a paired statistical test of the difference between workflows. The quality comparison should be read as descriptive.

Metric	Baseline	S-SPARC	Descriptive difference
Average token use per student	8,525.15	1,980.20	-6,544.95
Token reduction	–	77%	–
Average code-quality score	87.72%	88.13%	+0.41 percentage points
S-SPARC 95% confidence interval	–	88.13 ± 5.86	–
Paired token-use test	–	$p = 0.029$	significant
Paired token-use statistic	–	$t(19) = 2.36$	–
Effect size	–	Cohen’s $d = 0.53$	medium

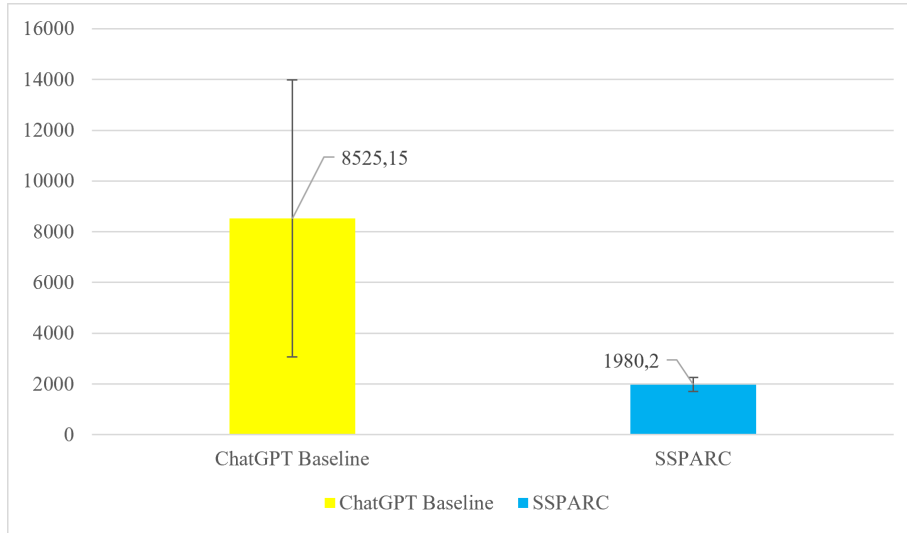


Figure 3: Average token use in the within-participant comparison. S-SPARC used fewer tokens than the baseline workflow; code-quality scores are reported separately and are not interpreted as statistically equivalent.

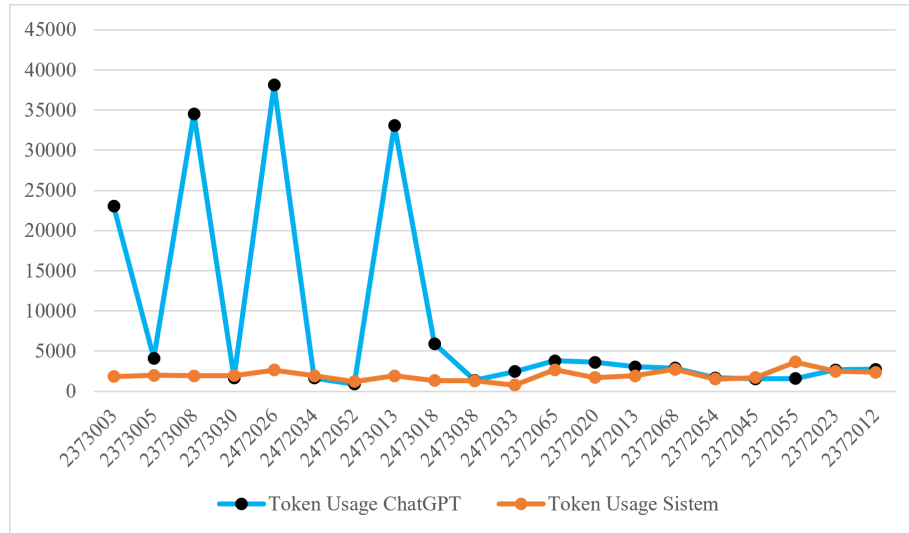


Figure 4: Token use per student in the within-participant comparison. The figure displays the participant-level values reported in the thesis appendix.

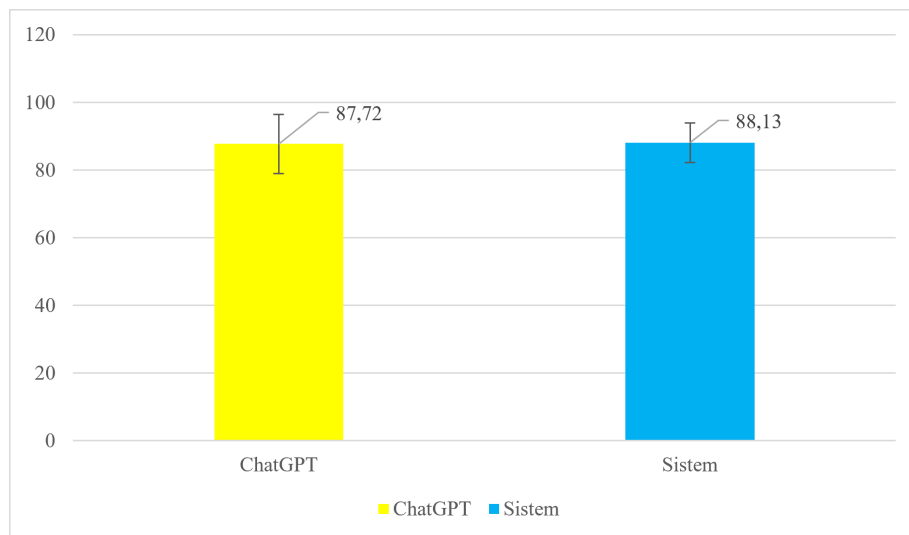


Figure 5: Descriptive code-quality scores for the baseline workflow and S-SPARC, with a reported difference of 0.41 percentage points. The 95% interval reported here describes uncertainty around the S-SPARC mean only; it is not a paired statistical test of the difference between workflows. The quality comparison should be read as descriptive.

5.3. Semester-long reduction and variation

This subsection reports the semester-long results for RQ1 and RQ2.

Potential token demand across seven laboratory sessions was 2,122,873 tokens. Retrieval handled 1,781,845 tokens, while 341,028 tokens were forwarded to external LLM inference. The cumulative reduction in external LLM inference was 83.94%.

Table 8: Token workload by laboratory session in the semester-long quasi-experiment

Session	Retrieval tokens	External-inference tokens	Potential tokens
1	209,529	39,163	248,692
2	142,626	21,408	164,034
3	302,631	64,010	366,641
4	189,730	40,558	230,288
5	135,416	25,941	161,357
6	219,202	40,278	259,480
7	582,711	109,670	692,381
Total	1,781,845	341,028	2,122,873

In the first reported regression, the inference-token ratio was not significantly associated with response quality ($R^2 = 0.003$, adjusted $R^2 = -0.013$, $\beta = -0.088$, $t = -0.410$, $p = 0.683$). In the second reported regression, prompt quality had a significant positive coefficient ($\beta = 0.358$, $t = 16.248$, $p < 0.001$), week had a significant negative coefficient ($\beta = -0.153$, $t = -2.776$, $p = 0.006$), and prompt-token count was not significant ($\beta = -0.00011$, $t = -0.091$, $p = 0.927$). The second model had $R^2 = 0.438$. The thesis cautions that the week coefficient may reflect changes in task difficulty and course material rather than direct deterioration of the system. These statistics are reproduced from the thesis summaries; the underlying interaction-level data were unavailable for reanalysis.

The mean reduction rates for the early and late phases were 0.858 and 0.869, with variances of 0.00432 and 0.00413. The reported comparison was not significant ($t = -1.105$, $p = 0.273$). Weekly repeated-measures ANOVA also found no significant variation ($F = 0.859$, $p = 0.527$). The Greenhouse–Geisser corrected value was $p = 0.452$, and the reported generalized effect size was $\eta_g^2 = 0.0217$. The available files do not preserve the ANOVA degrees

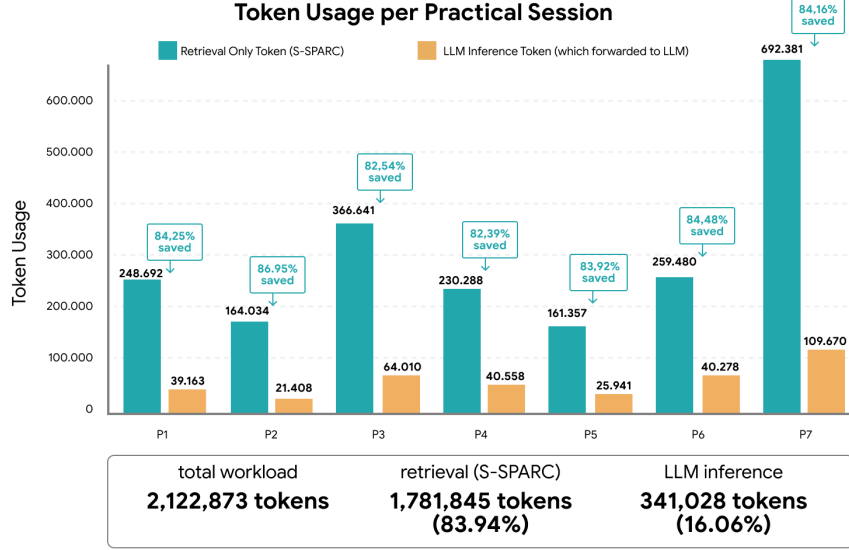


Figure 6: Cumulative token workload in the semester-long quasi-experiment. Retrieval handled most token demand, producing an 83.94% reduction in external LLM inference.

of freedom, correction coefficient, or nested raw observations. These non-significant tests do not establish equivalence or prove that the reduction rate was stable.

5.4. Environmental-footprint indicators

The tracker reported average per-execution indicators of approximately 0.000503 kWh, 0.0152 kg CO₂, and 0.19 L of water. For one controlled laboratory session, the thesis reports approximately 26.3 kg CO₂ and a carbon-derived electricity equivalence of approximately 34 kWh based on 0.8 kg CO₂ per kWh. These values are proxy indicators derived from token counts and literature-based energy estimates; they are not direct hardware measurements and cannot be independently reconciled with session-level totals.

No comparison of measured energy, carbon emissions, or water use between the baseline and S-SPARC workflows is documented. The reported session value should therefore not be interpreted as an avoided footprint or as an estimate of the environmental effect of the 77% token difference. The environmental API, conversion values, execution count, and aggregation steps required for independent verification are unavailable. The more directly

Table 9: Statistical summary of response quality and reduction-rate variation

Analysis	Statistic	Interpretation
Token-use difference, within-participant comparison	$p = 0.029$; Cohen's $d = 0.53$	Significant paired difference with medium effect
Inference-token ratio and response quality	$\beta = -0.088$; $p = 0.683$	No significant association in the reported model
Prompt quality coefficient	$\beta = 0.358$; $p < 0.001$	Significant positive coefficient
Week coefficient	$\beta = -0.153$; $p = 0.006$	Significant negative coefficient; not necessarily system decline
Early vs. late reduction rate	$t = -1.105$; $p = 0.273$	No difference detected; not an equivalence test
Weekly reduction-rate variation	$F = 0.859$; $p = 0.527$	No variation detected by the reported test
Greenhouse-Geisser correction	$p = 0.452$	Corrected test remained non-significant
Effect size for weekly variation	generalized $\eta_g^2 = 0.0217$	Small effect

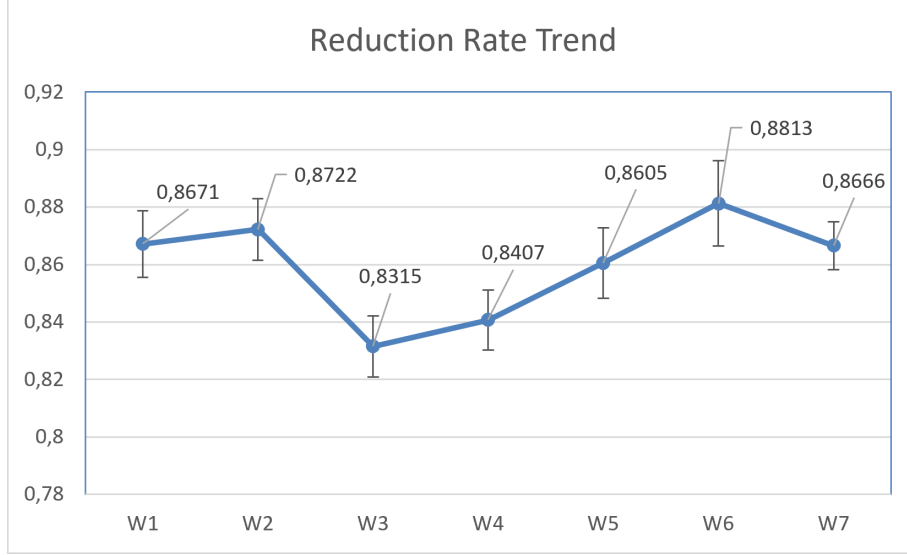


Figure 7: Weekly reduction rate across seven laboratory sessions. The figure describes observed variation and is not evidence of statistical equivalence across weeks.

supported resource result is the observed reduction in tokens sent to external LLM inference.

5.5. Survey observations

This subsection reports the available survey observations for RQ3.

Before the intervention, 60% of respondents reported using AI 6–15 times per week, 30% reported 0–5 uses, and 10% reported 16–30 uses. The thesis also reports that 50% used AI for 41–60% of their programming-task solutions and that 10% reported AI involvement in 61–80% of solutions. For code generation specifically, 50% reported using AI 3–5 times per week and 15% reported more than 10 uses per week.

The thesis presents pre- and post-use response distributions for three topics: the carbon-footprint concept, retrieval-oriented efficiency strategies, and the meaning of power usage effectiveness. The post-use distributions contain more correct or retrieval-oriented selections in the figures reported by the thesis. However, no documented matching procedure, inferential test, effect size, confidence interval, or validated scale is available. These patterns are therefore reported as differences between observed distributions, not as demonstrated within-person change or an effect caused by S-SPARC.

Table 10: Survey findings reported in the thesis

Survey result	Thesis-reported observation
Weekly AI-use frequency	60% used AI 6–15 times; 30% used it 0–5 times; 10% used it 16–30 times
AI involvement in programming solutions	50% reported 41–60% involvement; 10% reported 61–80% involvement
Weekly AI use for code generation	50% reported 3–5 uses; 15% reported more than 10 uses
Post-use carbon-footprint responses	The post-use distribution contains more correct selections; no within-person or inferential analysis is documented
Post-use efficiency-strategy responses	The post-use distribution contains more semantic-search and retrieval-oriented selections; no attribution test is documented
Post-use PUE responses	The post-use distribution contains more correct selections; no within-person or inferential analysis is documented

6. Discussion

6.1. Principal findings and educational implications

Together, the results address the three research questions within the limits of the available evidence. For RQ1, both the controlled comparison and the semester-long deployment recorded substantially less external LLM inference when validated retrieval was available. However, the controlled comparison evaluated complete workflows with different input capabilities, so the token difference cannot be attributed to retrieval alone. For RQ2, the reported mean code-quality scores differed by 0.41 percentage points, but no direct paired test or equivalence analysis was reported. The evidence therefore supports only a descriptive comparison, not a claim of equal or non-inferior quality. For RQ3, the survey documents frequent prior AI use and different pre- and post-use response distributions for several sustainability items. Because participant matching, identical measurement, inferential testing, and separate intervention conditions were not documented, the survey cannot demonstrate within-person change or attribute the observed distributions to footprint feedback, retrieval, or gamification.

S-SPARC shifts the focus of educational design from answer provision alone toward making resource use visible within the learning environment. This distinction is central to the paper’s novelty. The study demonstrates an implementable design in which validated reuse is presented before new generation and token-based footprint indicators are visible to learners. The available survey can describe participant responses after system use, but it cannot determine whether the design changed behavior, whether any change persisted, or which component contributed to the reported perceptions.

From an educational perspective, the study supports examining AI literacy beyond answer evaluation. Learners may also need to judge when generation is necessary, when validated knowledge can be reused, and how prompt quality relates to the support received. The reported association between prompt quality and response quality is consistent with this position, although it does not establish causation. In programming education, validated retrieval may provide a practical setting for discussing these judgments without presenting every interaction as a request for new generation.

From a sustainable AI perspective, the strongest evidence concerns token-based external inference demand rather than verified energy, carbon, or water savings. The environmental values cannot be independently reproduced because the underlying service and coefficients are unavailable. They should

therefore be used only as interface indicators. The more reproducible sustainability result is that retrieval handled 1,781,845 of 2,122,873 potential tokens during the seven-session deployment. This system-level measure shows how educational AI can reduce requests sent to an external model, while leaving the physical environmental consequences for future direct measurement.

The findings do not show that more generation produced better code-quality scores. However, they also do not prove that the two workflows were equivalent. A cautious interpretation is that the observed token reduction occurred alongside numerically close mean quality scores in this sample. The quasi-experiment further reported a positive prompt-quality coefficient, while the inference-token ratio was not significant in its model. These results justify further study of prompt quality and validated reuse, but not a general conclusion that reduced generation preserves learning outcomes in all settings.

6.2. Integrated design contribution

The architecture is intentionally integrated. Retrieval provides the mechanism for reducing external calls, validation limits which solutions can be reused, footprint feedback displays proxy indicators, and gamification provides an efficiency-oriented incentive. The present design shows how these components can be combined around responsible AI consumption. It does not establish the independent contribution of each component.

No formal ablation experiment was conducted. The retrieval statistics evaluate similarity-based reuse, but they do not test the added value of the validation, feedback, or gamification layers. Likewise, the survey reflects experience with the full system and cannot identify which component influenced a response. The integrated architecture is therefore a design contribution, while component-level effectiveness remains a question for future controlled studies.

6.3. Limitations, validity, and future research

This study has several reproducibility limitations. The available files do not preserve exact embedding-model revisions, software-library versions, OpenAI API parameters, the translation service, the static-analysis implementation, the final knowledge-admission cutoff, or the exact gamification penalty function. The documented formulas and operational retrieval threshold support partial reconstruction of the prototype, but not exact replication

of every system output. The 0.90 threshold was calibrated using the available MBPP and HumanEval benchmark data and was not validated on an independent held-out source.

The retrieval assessment does not document the number of human annotators, annotation rubric, inter-rater agreement, or the method used to aggregate 4,000 pairwise judgments into prompt-level outcomes. Threshold selection and evaluation used the same labeled sets grounded in MBPP’s 974 problems and HumanEval’s 164 problems, and the weak labels used embedding-based criteria related to the retrieval method. The reported retrieval values should therefore be interpreted as prototype calibration rather than independent validation, with generalization constrained by the size and domain coverage of those benchmark datasets.

Internal validity is affected by the quasi-experimental setting, the absence of a separate gamification or footprint-feedback condition, and possible familiarity with course tasks. The controlled comparison used the same participants and tasks in both workflows, but condition order and counterbalancing were not documented. Direct ChatGPT also accepted multimodal or full-description inputs, whereas S-SPARC processed relevant text. The token comparison consequently represents the implemented workflows rather than the isolated effect of retrieval. Code-quality assessment lacks documented raters, rubric anchors, blinding, and inter-rater reliability.

External validity is limited by the study context: programming education, Indonesian student interactions, English benchmark-derived knowledge, 20 students in the controlled comparison, and 55 students in one semester-long quasi-experiment. The translation component could not be evaluated independently because its service and version were not recorded. Replication across institutions, languages, and course levels is needed before the findings can be generalized more broadly.

Construct validity depends on using token use as a proxy for external inference demand and token/API-based outputs as environmental indicators. The environmental API and coefficients are unavailable, and the reported per-execution and session-level conversions cannot be independently reconciled. The manuscript therefore does not claim measured energy use or verified avoided emissions. The pre- and post-surveys were not identical, and no validated scale, reliability analysis, participant-matching procedure, or matched inferential test was documented. Their results are limited to the observed distributions and post-use perceptions reported in the thesis.

The thesis reports aggregate regression and repeated-measures results,

but the raw interaction-level observations, nesting structure, ANOVA degrees of freedom, and correction coefficient are not available in the project files. These analyses could not be independently reproduced or extended with multilevel models. Non-significant early-late and weekly tests do not demonstrate equivalence or long-term stability.

Future work should preserve versioned source code, model revisions, configuration files, raw interaction logs, annotation protocols, rater information, and analysis-ready data. It should also use a held-out retrieval test set, document condition order, apply blinded quality assessment with reliability reporting, and use validated matched survey measures. Direct hardware measurement would improve environmental validity. Controlled component comparisons and longer behavioral studies are needed to determine whether feedback and gamification produce transferable changes in AI use.

7. Conclusion

S-SPARC demonstrates a design in which reducing external LLM inference is made visible within programming education. In the controlled comparison, average token use was approximately 77% lower with S-SPARC, while the reported mean code-quality scores differed by 0.41 percentage points. Because no equivalence analysis was specified and the workflows differed in input capabilities, these findings should not be interpreted as proof of equal quality or as an isolated retrieval effect. In the semester-long quasi-experiment, retrieval handled 83.94% of cumulative potential token demand. The reported tests detected no early-late or weekly difference, but they do not establish equivalence or long-term stability.

The broader lesson is both educational and methodological. Educational AI systems can be evaluated not only by the answers they provide, but also by the external inference demand associated with their use. The conceptual contribution is to frame responsible AI consumption as selective use of generative assistance. The design contribution is to combine validated retrieval, knowledge validation, proxy footprint feedback, and gamified regulation around this objective. The empirical contribution is evidence of reduced external inference demand in the studied workflows, accompanied by descriptive quality and survey results. Direct environmental measurement, component-level causal tests, and independent replication remain necessary.

Ethics statement

The thesis and project files available for this manuscript do not contain an ethics-committee approval identifier or a documented consent procedure. The manuscript therefore does not claim formal ethics approval or documented informed consent. These records should be verified and supplied before publication if they exist outside the available project materials.

Data and code availability

Participant-level values for the controlled comparison and respondent-level survey responses are available in the thesis appendices. The project files available for this manuscript do not include versioned source code, raw system logs, machine-readable retrieval judgments, environmental-API configuration, or interaction-level data from the semester-long deployment. Exact computational replication is therefore not currently possible from the available materials.

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